INCH-POUND

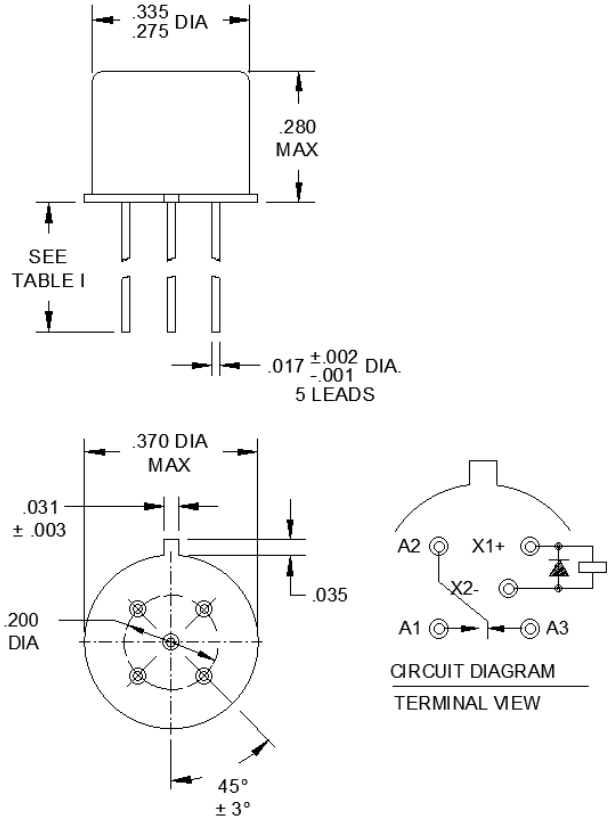
MIL-PRF-39016/23G
14 March 2022
SUPERSEDING
MIL-PRF-39016/23F
15 June 2005

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, SPDT, LOW LEVEL TO 1.0 AMPERE WITH INTERNAL DIODE FOR COIL TRANSIENT SUPPRESSION

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the relays described herein shall consist of this specification sheet and the latest issue of MIL-PRF-39016.



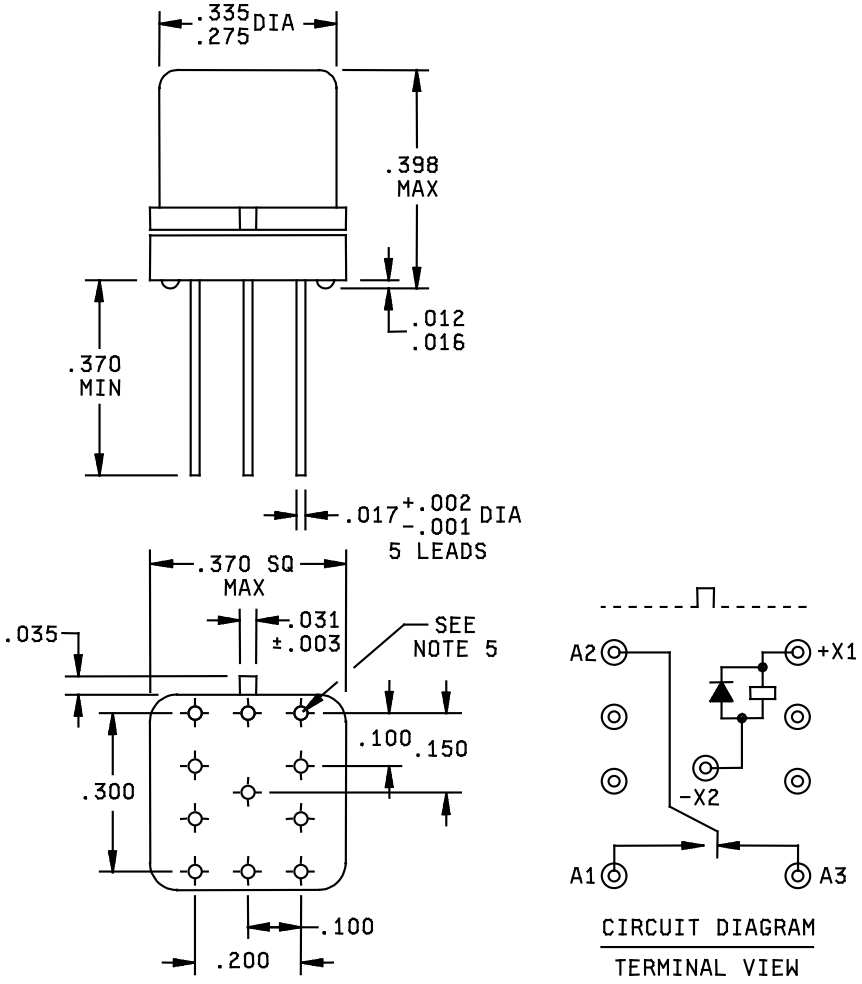
Inches	mm
.001	0.03
.002	0.05
.003	0.08
.017	0.43
.031	0.79
.035	0.89
.200	5.08
.275	6.99
.280	7.11
.335	8.51
.370	9.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above are for reference only. Numbers do not appear on the relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. All leads shall be electrically insulated from the case.
7. Coil symbols are optional per MIL-STD-1285.
8. Circuit diagram shown on part is the terminal view.

FIGURE 1. Dimensions and Configuration.

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Inches	mm
.001	0.03
.002	0.05
.003	0.08
.012	0.30
.016	0.41
.017	0.43
.031	0.79
.035	0.89
.100	2.54
.150	3.81
.200	5.08
.275	6.99
.300	7.62
.335	8.51
.370	9.40
.398	10.11

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Spreader mounting pads shall be certified to A-A-55485, A-A-55485/5 (A-55485/05-004).
5. Dimensions and tolerance shown for the bottom view of the spreader mounting pad are for the center-to-center locations of the holes in the spreader mounting pad.
6. Shape optional within the envelope dimension.
7. Terminal numbers shown above for reference only. Numbers do not appear on relay.
8. Relays shall have a (+) sign placed on circuit diagram as shown.
9. All leads shall be electrically insulated from the case.
10. Coil symbols are optional per MIL-STD-1285.
11. Circuit diagram shown on part is the terminal view.

FIGURE 2. Dimensions and configuration relay with spreader mounting pad attached.

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REQUIREMENTS:

CONTACT DATA:

Load ratings:

High level (relay case grounded):

Resistive:

1.0 ampere at 28 V dc.
250 milliamperes at 115 V ac 60 and 400 Hz case not grounded.
100 milliamperes at 115 V ac 60 and 400 Hz case grounded.

Inductive load: 0.2 ampere at 28 V dc with 0.32 henry inductance.

Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 μA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 0.10 ohm maximum (0.125 ohm maximum with spreader mounting pad attached).

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.20 ohm maximum (0.225 ohm maximum with spreader mounting pad attached).

Low level:

During life: 33 ohms maximum.

After life: 0.15 ohm maximum (0.175 ohm maximum with spreader mounting pad attached).

Intermediate current:

During: 1 ohm maximum.

After: 0.20 ohm maximum (0.225 ohm maximum with spreader mounting pad attached).

Contact bounce: 1.5 milliseconds maximum (applicable to failure rate level “L”).

Contact stabilization time: 2.0 milliseconds maximum (applicable to failure rate levels “M”, “P”, and “R”).

Overload (high level only): Two times rated current. Not applicable to ac load ratings.

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COIL DATA: See table I.

Operate time: 2.0 ms maximum over temperature range with rated coil voltage.
Release time: 4.0 ms maximum over temperature range from rated coil voltage.

ELECTRICAL DATA:

Insulation resistance: 10,000 megohms minimum at 500 V dc, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage:	Sea level V rms (60 Hz)	Post intermediate current life test Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts in the energized and de-energized positions.	500	500	
Between case, frame, or enclosure and coils.	500	500	300
Between all contacts and coils.	500	500	All terminals to case
Between open contacts in the energized and de-energized positions.	500	375	
Between contact poles.	N/A	N/A	
Between coils of dual coil relays.	N/A	N/A	

DIODE CHARACTERISTICS: 1/

Maximum negative transient: 1 volt.

Coil transient suppression: Applicable.

Semiconductor in-process screening: Applicable, visual inspection of semiconductors shall be in accordance with MIL-STD-750, method 2073 or 2074.

ENVIRONMENTAL DATA:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): MIL-STD-202-204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Vibration (random): MIL-STD-202-214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts. Applicable to qualification and group C testing only.

Shock (specified pulse): MIL-STD-202-213, test condition B (75 g's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

1/ WARNING: Reverse polarity on coil terminals will destroy diode.

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Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): In accordance with MIL-STD-750, method 1041.

PHYSICAL DATA:

Terminal strength: MIL-STD-202-211

Pull test: Test condition A, 1 pound pull.

Bend test: Test condition C, ½ pound load.

Twist test: As specified in MIL-PRF-39016.

Solderability: Applicable.

Dimensions and configuration: See figures 1 and 2.

Weight: 2.27 grams (0.08 ounce) maximum, 2.52 grams (0.089 ounce) maximum with spreader mounting pad attached.

Seal: Hermetic.

Minimum marking: Military part number, “J” with the date code (example J0430), circuit diagram, manufacturer’s name or source code.

LIFE TEST REQUIREMENTS:

High level: 100,000 cycles per relay.

Low level: 100,000 cycles plus 900,000 cycles mechanical life.

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TABLE I. Dash numbers and characteristics. 1/

Dash numbers 2/				Coil Voltage (V dc) 3/		At 25°C				Over temperature range		
Lead length 1.500 min 4/	Lead length .187 ±.010	Lead length .500 min	Relays with spreader mounting pads (fig.2) 5/	Rated	Max	Coil resistance ohms ±10%	Specified pick-up value (volt.) (V dc)	Specified hold value (volt.) (V dc)	Specified drop-out value (volt.) (V dc)	Specified pickup value (volt.) (V dc)	Specified hold value (volt.) (V dc)	Specified drop-out value (volt.) (V dc)
013	019	025	031	5.0	5.8	63	2.8	1.7	0.23	3.7	2.4	0.15
014	020	026	032	6.0	8.0	125	3.5	2.0	0.28	4.5	2.8	0.18
015	021	027	033	9.0	12	280	5.3	3.0	0.54	6.8	4.2	0.35
016	022	028	034	12	16	500	7.0	4.0	0.63	9.0	5.6	0.40
017	023	029	035	18	24	1,130	10.5	6.0	0.91	13.5	8.4	0.58
018	024	030	036	26.5	32	2,000	14.2	8.0	1.37	18.0	10.4	0.89

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits not recommended for subsequent use in low level applications.
2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example, 013L - - - - - -030R.
3/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.
4/ 1.500 leads are inactive for new design.
5/ Relay supplied with spreader mounting pads (-031 through -036) shall have the spreader mounting pad rigidly attached.

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QUALIFICATION INSPECTION:

Qualification inspection and sample size: See table II.

TABLE II. Qualification inspection and sample size. 1/

Single submission	Group	submission
18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable	M39016/23-030	18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable
	M39016/23-025 M39016/23-026 M39016/23-027 M39016/23-028 M39016/23-029	2 units each part number Qualification inspection, Q1.

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-PRF-39016/15, /16, /20, /21, /24, /25, and /26 may be used in addition to MIL-PRF-39016/23 data. Prior to performance of retention of qualification testing, the relay manufacturer shall preselect the sampling plan.

2/ The number of units required for qualification testing shall be increased as required in Q5, MIL-PRF-39016, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to performance of qualification inspection, the relay manufacturer shall preselect the sampling plan.

Initial qualification of relays supplied with spreader mounting pads (-031 through -036), shall be tested as specified below:

Perform the following tests as specified in the qualification inspection table of MIL-PRF-39016, in the order shown below:

Before installation of spreader mounting pad, screening, visual and mechanical examination (internal), thermal shock, resistance to solvents, vibration (sinusoidal), vibration (random), shock (specified pulse), acceleration, terminal strength, magnetic interference (when specified), capacitance (when specified), coil life (applicable to continuous duty relays only), resistance to soldering heat, salt spray (corrosion), overload (applicable to high level relays only), life, terminal strength, and intermediate current.

After installation of spreader mounting pad perform the following tests as specified in the qualification inspection table of MIL-PRF-39016, in the order shown below:

Insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

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Qualification inspection (reduced testing for previously qualified relays) for relays supplied with spreader mounting pads (-031 through -036), two (2) units of the 26.5 volt rated coil voltage (-036) shall be tested as specified below:

Before installation of spreader mounting pad perform the following tests as specified in the qualification inspection table of MIL-PRF-39016 in the order shown below:

For failure rate level L only: Screening.

For failure rate levels M, P, and R: Vibration (sinusoidal) test duration shall be 10 minutes, vibration (random), and screening.

After installation of spreader mounting pad perform the following tests as specified in the qualification inspection table of MIL-PRF-39016 in the order shown below:

Insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

Group A testing for relays supplied with spreader mounting pads (-031 through -036), shall be tested as specified below:

Perform seal test immediately, preceding the A2 electrical tests. Relay leads shall be formed and the spreader mounting pad removed before the seal test. After the seal test, the spreader mounting pad shall be rigidly attached to the relay and the remaining group A tests performed.

Qualification inspection (reduced testing) and sample size: See table III.

If the relays produced for MIL-PRF-39016/23 are similar in construction and design except for the diodes, coil assembly, and headers, as applicable to the relays produced for MIL-PRF-39016/15, /16, /20, /21, /24, /25, and /26, then reduced testing for qualification of MIL-PRF-39016/23 relays may be performed concurrent with or subsequent to successful qualification of MIL-PRF-39016/15, /16, /20, /21, /24, /25, and /26. For reduced testing, see table III.

TABLE III. Qualification inspection (reduced testing).

Examination or test
2 units each coil voltage – Q1 of qualification inspection table 1 unsealed sample unit for internal examination.

SUPERSESSION DATA:

Supersession data: See table IV.

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TABLE IV. Supersession data. 1/

Superseded part no. M39016/23	New part no. M39016/23- 1/	Superseded part no. M39016/23-	New part no. M39016/23-
001	013	007	019
002	014	008	020
003	015	009	021
004	016	010	022
005	017	011	023
006	018	012	024

1/ Dash numbers -013 through -018 are inactive for new design and are for support of existing equipment designs only.

Cross reference for Government logistical support: See table V.

TABLE V. Cross reference for Government logistical support.

Superseded part Number M39016/23-	New part number M39016/23-	Support with part number M39016/	New part number M39016/23-	Support with part number M39016/
001	013	23-013	025	23-025
002	014	23-014	026	23-026
003	015	24-015	027	24-027
004	016	24-016	028	24-028
005	017	24-017	029	24-029
006	018	24-018	030	24-030
007	019	23-025	031	23-031
008	020	23-026	032	23-032
009	021	23-027	033	24-033
010	022	24-028	034	24-034
011	023	24-029	035	24-035
012	024	24-030	036	24-036

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Referenced documents. In addition to MIL-PRF-39016, this document references the following:

A-A-55485 A-A-55485/5 MIL-PRF-39016/15 MIL-PRF-39016/16 MIL-PRF-39016 /20 MIL-PRF-39016/21 MIL-PRF-39016 /24 MIL-PRF-39016 /25 MIL-PRF-39016/26 MIL-STD-1285 MIL-STD-750 MIL-STD-202-204 MIL-STD-202-214 MIL-STD-202-213 MIL-STD-202-211

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Custodians:
Army - CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2022-020)

Review activities:
Army – AR
Navy – AS, MC, OS, SH

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